

JIBIN BENNY

linkedin.com/in/jibin-benny | +91 7559979727 | jibinbenny06@gmail.com | github.com/jibin006

SUMMARY

Cloud Security Engineer specializing in build-focused security across Google Cloud, AWS, and Azure. Designs and implements Python security automation, policy-as-code pipelines (Terraform/OPA/Conftest), CI/CD security gates (SAST/DAST/IaC), and container hardening for Docker and Kubernetes (GKE/EKS). Building AI/ML security controls including prompt injection detection, RAG indirect injection defense, inference safety testing, and LLM threat evaluation aligned with OWASP Top 10 for LLMs. GCP Professional Cloud Security Engineer certified with hands-on Workload Identity Federation, CMEK/KMS, VPC Service Controls, and private networking.

EXPERIENCE

Cloud Security Engineer | Equifax | Trivandrum, India

07/2024 – Present

- Architected secure-by-default multi-account AWS landing zone with reusable Terraform modules enforcing IAM permission boundaries, private networking with VPC endpoints, CMEK encryption defaults, and tagging guardrails across 3 OU layers and 6+ SCPs — eliminating privilege-escalation and misconfiguration patterns before they reach production, reducing blast radius of a compromised account to a single OU boundary.
- Implemented 14 OPA/Conftest policy-as-code gates with progressive rollout (audit → warn → enforce) alongside SAST/IaC scanning (tfsec, Checkov) — blocking wildcard IAM actions, open security groups, and unencrypted resources at PR-time, shifting security left and eliminating class of misconfigurations from reaching deployment pipelines.
- Redesigned CI/CD identity model using OIDC federation and Workload Identity with 5 scoped trust policy conditions, eliminating long-lived credentials and reducing credential exfiltration attack surface to zero for automated pipelines. Threat modeled 4 compromise scenarios: compromised runners, malicious PRs, dependency confusion, direct registry push.
- Established 3-layer supply chain trust: Cosign artifact signing, SBOM generation + vulnerability scanning (Syft/Grype/Trivy), and Kubernetes admission control blocking unsigned artifacts from production — preventing tampered images from executing in cluster.
- Built 3-tier detection and response pipeline (auto-remediate / enrich-alert / suppress) integrating GuardDuty, Security Hub, CloudTrail with EventBridge-driven Python Lambda — reducing mean time to contain credential abuse events by automating IAM key revocation and instance isolation without manual intervention.
- Built three-layer LLM inference security pipeline in Python defending against authorized misuse beyond infrastructure controls: Model Armor baseline, custom detector (pattern matching, structural heuristics, RAG indirect injection scanning via `analyze_context()`), and PII output filter — fail-closed by design. Measured across 38 test cases: 100% detection rate, 4.5% FP rate, 0.19ms avg latency. Prevents model inversion, prompt extraction, and data boundary violations in production inference endpoints.

Cloud & Network Security Engineer | Tata Consultancy Services | Kochi, India

04/2021 – 07/2024

- Engineered secure multi-cloud network architectures (AWS/Azure) with 4-layer segmentation — VPC/VNet isolation, subnet tiers, security group rules, routing controls, and VPC endpoints — reducing lateral movement paths across client environments at scale.
- Built Python security automation (boto3) for IAM drift detection with privilege escalation scoring and misconfiguration analysis, shifting detection from periodic manual review to near real-time alerting across multi-cloud environments.

CORE EXPERTISE

Cloud Security: GCP (Workload Identity, Org Policies, VPC Service Controls, CMEK, Vertex AI Security, Private Networking) | AWS (IAM, SCPs, Permission Boundaries, OIDC Federation, GuardDuty, Security Hub, CloudTrail, KMS) | Azure

AI/ML Security: OWASP Top 10 for LLMs | MITRE ATLAS | Prompt Injection Defense (Direct + Indirect RAG) | Inference Safety Testing | PII Output Filtering | Agentic AI Threat Modeling | Model Armor | Fail-Closed Design

Policy-as-Code & IaC: Terraform | OPA/Rego | Conftest | tfsec | Checkov | CI/CD Security Gates | SAST/DAST/SCA

Container & Kubernetes: GKE/EKS | Gatekeeper | RBAC | Network Policies | Pod Security Standards | Falco | Cosign/Sigstore | SBOM (Syft/Grype)

Detection & IR: Detection Engineering | EventBridge Automation | GuardDuty | CloudTrail Lake | Incident Response | RCA | Containment Automation

Automation: Python (boto3, FastAPI) | Lambda | Bash | Step Functions | SQL (PostgreSQL)

PROJECTS & OPEN-SOURCE

LLM Security Framework | Python, FastAPI

Three-layer prompt injection detection pipeline: Model Armor baseline, custom Python detector (13 rules across 7 OWASP LLM categories, 4 structural heuristics, analyze_context() for RAG indirect injection), and PII output filter (7 patterns, system prompt leak detection). Fail-closed FastAPI proxy with /detect, /scan-context, /safe-inference endpoints. Defends against model inversion, data exfiltration via RAG, and jailbreak-driven boundary violations. Measured: 100% detection, 4.5% FP rate, 0.19ms avg latency, 0.88ms P95 across 38 test cases.

GCP Secure AI/ML Infrastructure | Terraform, GCP

Workload Identity Federation (keyless auth), GCP Org Policies, VPC Service Controls, CMEK encryption, Cloud Audit Logs to BigQuery, and Terraform module for secure Vertex AI deployment with private endpoints and model registry IAM controls. OPA policy validation in CI. Designed to prevent model exfiltration and lateral movement from compromised training pipelines.

IAM Drift Detector | Python, Lambda

Cross-account IAM drift detection using differential snapshot analysis with 4-level severity scoring based on blast radius (wildcard permissions, trust policy changes, escalation paths). Automated Lambda + EventBridge pipeline. Validated against 9 escalation scenarios.

Terraform Secure Modules + OPA Policies | Terraform, Rego

4 hardened modules (S3, IAM, VPC, Lambda) with enforced security defaults + 14 OPA/Conftest policies blocking misconfigurations at PR-time. Policy testing framework with GitHub Actions CI integration.

Kubernetes Security Enforcement | Kubernetes, Falco

EKS/GKE stack: RBAC namespace isolation, Pod Security Standards, default-deny network policies, 18 Gatekeeper constraints, Docker hardening, Cosign admission verification, and 5 Falco runtime detection rules with runbooks.

CERTIFICATIONS

- GCP Professional Cloud Security Engineer (2025)
- AWS Security Specialty — In Progress, Expected: 06/2026
- CKS – Certified Kubernetes Security Specialist — In Progress, Expected: 08/2026

EDUCATION

B.Tech, Electrical and Electronics Engineering | Kerala Technological University | 2020